A HUMAN CORNEA-ON-A-CHIP

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ABSTRACT

This paper describes a microfluidic platform to reconstitute human cornea and the complexity of its interface to external environment. To recapitulate the structural complexity and biological phenotype of the cornea, we embedded a collagen-coated porous membrane in a double-layer microfluidic chip for three-dimensional (3D) culture of cells derived from the human cornea. Moreover, a pressure-driven flow was used for long-term cultivation, which not only forms a bionic human cornea, but also simulates the microenvironment of the human ocular surface. We demonstrated a novel micro-engineered device that would improve the interaction between the *in vitro* model and the external environment, providing a potential solution to obtain predictive results to *in vivo* observations.

KEYWORDS

Microfluidics, Organ-on-a-chip, Biomimetics, Eye, Cornea, Ocular surface, Cell culture, 3D culture

INTRODUCTION

Human eye is the visual organ and the most important organ in the human senses. As a principal protective barrier in this unit, the cornea consists of closely apposed epithelium and endothelium separated by a collagen-rich stromal tissue that contains keratocytes, and the ocular surface is also continuously affected by the dynamic microenvironment [1].

However, the structural, functional and environmental complexity of the ocular surface poses major technical challenges for *in vitro* studies of its physiology and pathology using traditional cell culture models. The *in vitro* studies are usually performed under static conditions that are different from *in vivo* dynamic conditions, i.e. eye blinking and tear secretion [2]. Therefore, research in this field has relies on controversial and time-consuming *in vivo* animal studies, which usually fail to model biological responses in humans [3-5]. While the animal models can reflect the overall situation of the organism, studies have shown the limited ability of these tests to recapitulate the complex structure of human cornea due to species differences. For example, the low blinking rate in rabbits can cause higher interaction between the drug and the ocular surface, resulting in higher drug permeation, which may mislead preclinical test results [6].

These critical drawbacks of exiting models have limited our basic understanding and thus hindered the development of new treatments for ocular diseases. To solve the serious lack of *in vitro* eye models, we have developed a cornea-chip based on microfluidic technology, which replicates 3D structure, physiological function and dynamic microenvironment of the human cornea.

EXPERIMENTAL

Cornea-chip Design and Fabrication

Cornea is the outermost organ structure of eye, which consists of epithelium, Bowman's layer, stroma, Descemet's layer, and endothelium (Figure 1A). In this work, we described a biomimetic human corneal model on a microfluidic device with an integrated 0.4 μm collagen-coated porous polycarbonate (PC) membrane and polydimethylsiloxane (PDMS) microfluidic channels. A well structure with an open top was constructed on the upper layer of the device made by PDMS and the configuration of the microfluidic device was designed as that the upper and the lower microfluidic channels can be controlled independently. Then, immortalized human corneal epithelial (HCEpi) cells and immortalized human corneal endothelial (HCEnd) cells were seeded on the upper and lower side of the membrane, respectively, and cultured in nutrients that were injected into different microchannels (Figure 1B). A hole for Trans-epithelial electrical resistance (TEER) measurements in situ was also made on the device and connected to the cell culture area that was located at the middle of the microfluidic channels (Figure 1D). The open top of the cell culture area provides an air-liquid interface for culturing the corneal epithelium, so the physiological characteristics of the device are similar to the human cornea.

The microfluidic device was fabricated by directly bonding a glass substrate with two PDMS layers sequentially, and an extracellular matrix (ECM)-coated PC membrane was sandwiched in between (Figure 1C). The upper and lower PDMS layers with microchannels were both replicated from SU-8 structures that were fabricated by soft lithography. A 10:1 (w/w) mixture of PDMS prepolymer and curing agent was cast over the SU-8 molds and cured at 80 °C overnight. Then the PDMS replicas were peeled off from the molds, resulting in microfluidic channels with a height of 220 μm. Access holes for inlets and outlets of the microfluidic channels were punched on the PDMS slabs. At the middle of the microfluidic channel, a circular hole with diameter of 6 mm was punched on each PDMS slab. This was used to mimic the human corneal structure, and the open top surface of this area was used to create an air-liquid interface and mimic the ocular surface more precisely. By completely cleaning the glass substrate and the PDMS layers, they were then bonded together by oxygen plasma with the PC membrane embedded between the two microfluidic channels. Details on the microfabrication procedure are presented in Figure 2. Scanning electron microscopy images showing the cross-section of the culture region is shown in Figure 3.

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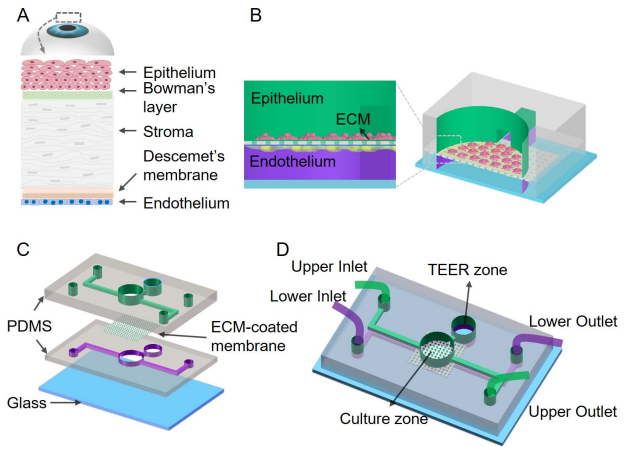

Figure 1. *Schematic diagram of the human cornea and design for the cornea-chip. (A) The anatomy of the human cornea. (B) Cross-sectional schematic diagram of the corneal chip. HCEpi cells and HCEnd cells were cultured on opposite sides of the membrane, the ECM-coated porous membrane was embedded in between the two PDMS layers. (C) 3D illustration on the device, showing the details on the structures on different layers. (D) Photograph of the human cornea-chip during verification with dyes of the two fluidic channels. The circular hole was at the middle ("Culture zone") to mimic the human corneal structure. Structures for in situ TEER measurements ("TEER zone") are also fabricated on the chip.*

Human Cornea-chip Construction and Characterization

Prior to the surface treatment, the chip was sterilized by UV germicidal irradiation in a biosafety cabinet for 30 minutes. Then, collagen (0.05 mg/mL in 0.02 M acetic acid) was used as the ECM to treat the PC membrane at 37 °C for 2 hours [7,8]. Excess proteins were then aspirated and the membrane allowed to air dry. HCEpi cells and HCEnd cells were introduced into the device at a concentration of 3×10^6 cells/mL and 5×10^5 cells/mL through the upper and the lower microchannel, respectively. The cells were then seeded on each side of the membrane and cultured by their own medium at a continuous flow condition with volumetric flow rate of 100 μL/h by using a programmable syringe pump, and the device was kept in a CO_2 incubator. Once the epithelial layer was confluent on the membrane, the culture medium was gently aspirated from the 6 mm hole at the middle of the upper microfluidic channel, air was then filled the epithelial compartment to create an air-liquid interface and mimic the ocular surface. As a comparison, the HCEpi cells were also cultured in a transwell chamber with an air-liquid interface and details are described in Figure 4. TEER values were measured in situ using a Millicell-ERS electrical resistance system at different days to evaluate barrier integrity of the corneal epithelium. After 14 days, the cornea-chip was characterized using immunofluorescence staining and monitored by a Zeiss Axio Observer 7 inverted fluorescence microscope equipped with light-emitting diodes, appropriate fluorescent filter sets and a sCMOS camera (Hamamatsu ORCA-Flash4.0-V3).

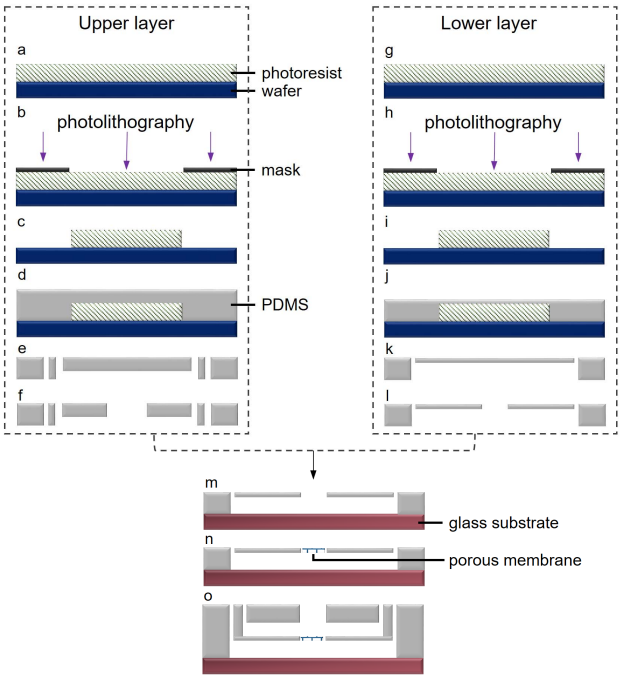

Figure 2. *Schematic illustration on the microfabrication of the cornea-chip. (a, g) SU8 photoresist was spin-coated on the silicon wafers and (b, h) patterned by UV exposure and (c, i) developed. (d, j) PDMS was poured onto the SU8 molds. (e, k) The PDMS slabs were peeled off and punched to make access holes. (f, l) The PDMS slabs were further punched to make open wells. (m) The lower PDMS slab and the glass substrate are bonded by oxygen plasma treatment. (n) The open well in the lower PDMS slab was sealed by a porous polycarbonate membrane. (o) The upper PDMS slab was bonded to lower layer to create a multilayered cornea-chip.*

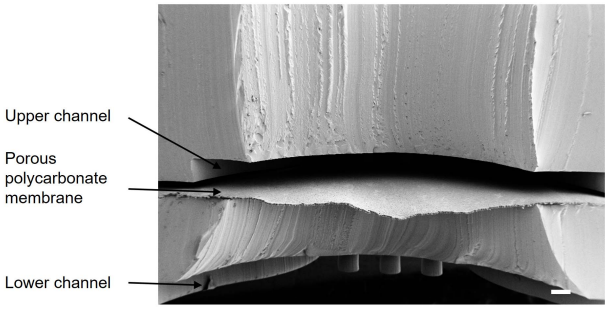

Figure 3. *Scanning electron microscopic image on the cross-section of the microfluidic device over the cell culture region. Scale bar, 200 μm.*

Statistical analysis

All results were presented as mean ± standard deviation (S.D.). The unpaired t-test or a two-way analysis of variance (ANOVA) followed by Bonferroni's multiple comparisons test was performed using GraphPad Prism 8. The experiments were repeated and P values are obtained from source data and reported in figure legends.

RESLUTS AND DISCUSSION

To investigate whether the chip was suitable for human corneal cells culture, the cells were maintained in

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the microfluidic device for 14 days in the following experimental groups: on a transwell chamber ("Control" in Figure 4A) and on the ECM-coated PC membrane with an air-liquid interface on-chip ("Cornea-chip"), respectively. The cells were stained by Calcein-AM at the end of 14 days, and the cell viability from these two groups was compared, as shown in Figure 4A. For the construction of the biomimetic human corneal tissue in our device, the seeded epithelial cells were first grown to confluence in submerged conditions and then cultured at the air-liquid interface to mimic their native organ-specific environment. The HCEpi cells cultured on the microfluidic chip formed a stratified epithelium consisting of three to five stacked cell layers, which recapitulated the corneal epithelium *in vivo* (Figure 4B). The TEER was measured between the HCEpi and the HCEnd layer to evaluate barrier integrity after culturing cells for 3, 7, 10, and 14 days, respectively (Figure 4C). The measurement results were presented as $\Omega \cdot cm^2$ by multiplying the resistance with the surface area ($0.28\ cm^2$) of the open well. The resistance of the unseeded cell membrane was subtracted from all readings before calculations. The human corneal epithelium began to form a barrier function when the cells were cultured under the air-liquid interface condition for 3 days. After cultured for a long time, the barrier function obtained from the cornea-chip was higher than that from the transwell model (Figure 4D).

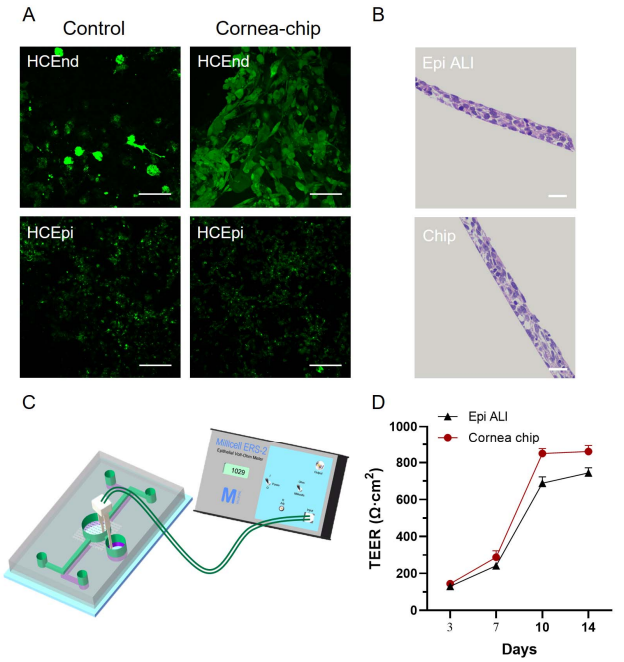

Figure 4. *Construction of physiological phenotypes of the human cornea in the device. A. Microscopy images of the human epithelial and endothelial cells viability in different groups. Control, Green, Calcein-AM. Scale bars, 50 µm. B. Representative hematoxylin and eosin (H&E) images of the stratified corneal epithelium for 14 days. Scale bars, 20 µm. C. Schematic diagram of on-site TEER measurement. D. TEER values of the Epi only culture at ALI and the cornea-chip. All data were repeated three independent experiments to ensure consistent results.*

The epithelial and endothelial tight junctional barrier plays a key role in homeostasis and host defense of cornea [9]. The expression of tight junction proteins can reflect the barrier function. A substantial increase on the expression of zona occludens-1 (ZO-1) protein accompanies with the full barrier function. The HCEnd or HCEpi cells were separately cultured in the chip as the control group and the above cells were cultured in the cornea-chip with an air-liquid interface as the experiment group. The cell morphology and the expression of ZO-1 of HCEnd and HCEpi cells are shown in Figure 5. After 14 days, the morphology of the cells remained very well and the expression of ZO-1 was significantly increased on both HCEnd and HCEpi cells that were cultured on the chip (Figure 5A, C). Interestingly, it should be noted that the ZO-1 expression of the cells cultured on cornea-chip for 7 days was higher than that control group, the value kept increasing with time but became a bit lower than control group after 14 days (Figure 5B, D). This indicates that contact inhibition may occurs when the two cells were co-cultured from 7 to 14 day, which is more similar to the situation *in vivo* [10].

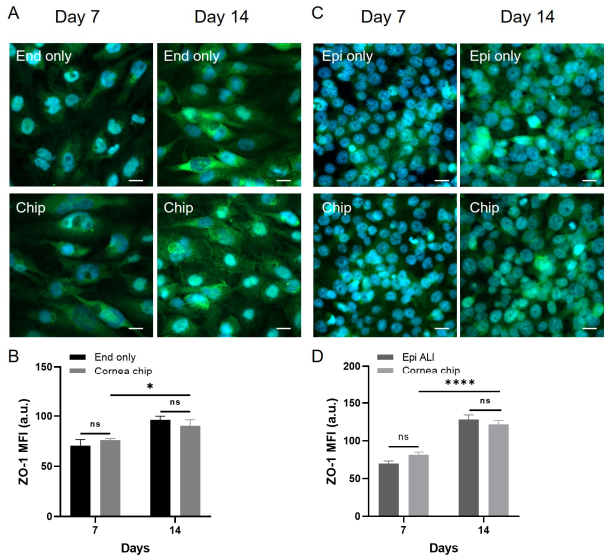

Figure 5. *Immunofluorescent characterization of HCEnd and HCEpi cells cultured in different conditions. (A) HCEnd cells are cultured either in a fully submerged condition (End only) or in cornea-chip (Chip). (B) Mean fluorescent intensity of ZO-1 protein expression obtained from different groups in (A). (C) HCEpi cells are cultured either in chip (Epi only) or in cornea-chip with air-liquid interface (Chip). (D) Mean fluorescent intensity of ZO-1 protein expression obtained from different groups in (C). (A, C) Green color is the immunofluorescence staining of ZO-1 protein and blue color shows the cell nucleus by DAPI staining. Scale bar, 20 µm. (B, D) MFI, mean fluorescence intensity. a.u., arbitrary unit. Data are mean ± S.D. of relatively fluorescence intensity from three independent experiments. P values by Two-way ANOVA and Bonferroni's multiple comparisons test. "ns" no significant difference, "*" P<0.05, "**" P<0.01, "***" P<0.005, "****" P<0.001.*

CONCLUSION

We developed a human cornea-on-a-chip using

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microfluidic technology in this study and validated its barrier effects. In brief, the cornea-chip has a complete human corneal structure, and its physiological characteristics is much closer to that of human subjects, so its suitability on the applications such as corneal disease study, evaluation of ocular drug absorption and permeation, as well as emulation of tissue barrier function and so on, can be envisaged in future. This corneal chip may replace animal models for cornea-related testing, which will provide meaningful, predictive and complimentary results to *in vivo* and *ex vivo* observations. Therefore, we believe that our technique could evolve in a tool for a wide class of biological and clinical applications.

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